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PERSONAL STATEMENT

My decision to pursue a Master of Engineering (MEng) degree at the University of British Columbia is driven by my commitment to advancing my technical capabilities in embedded systems and power electronics. With over five years of industry experience in motion control, automation, and electronics as an Engineer-in-Training, I am excited to expand and deepen my knowledge through UBC's rigorous, hands-on program. The MEng program aligns perfectly with my goals: to deepen my expertise, expand my system-level thinking, transition into advanced engineering roles, and take on opportunities to mentor and lead within the engineering community.

My technical foundation was built during my undergraduate studies at the British Columbia Institute of Technology, where I graduated with a bachelor's degree in electrical engineering. Courses in control systems, microcontrollers, and circuit design sparked my interest in electronics and embedded systems. My capstone project, developing an autonomous vineyard harvester, challenged me to integrate hardware and software in a real-world application. Our solution combined computer vision, GPS-based navigation, embedded microcontroller systems, and robotics for vine grape detection and picking. The project was a multidisciplinary effort that brought software, electronics, and mechanical design together. This experience opened my eyes to the interconnected nature of real-world engineering and sparked my passion for embedded systems and automation. To strengthen my foundation even further, I have also completed several online courses in areas such as embedded C programming and power supply design, which have helped me stay current with evolving industry practices and prepare for graduate-level study.

Professionally, I have worked across several sectors, including food and beverage automation, industrial energy systems, and electronics manufacturing. In my current role at Delta Intelligent Building Technologies (Canada) Inc, I design and program automated test fixtures for HVAC controller and thermostat PCBAs. This includes schematic design-for-test reviews, embedded test firmware development, mechanical fixture design, and production support, offering me practical insight into test automation, hardware-software interfacing, and manufacturing workflows. At Apex Motion Control, I integrated robotic arms and motion controllers to build custom automation solutions for food and beverage clients in North America and Asia. Through these experiences, I gained exposure to larger systems and recognized the critical need for deeper technical knowledge in embedded software, system integration, and resource-efficient system design.

As I look to the future, I am especially motivated to further develop my understanding of embedded system architecture, signal processing, and power electronics - fields that are increasingly critical in automation, robotics, and Internet-of-Things (IoT) systems. In four years, I envision myself working in an advanced design and management role, applying the knowledge and leadership skills gained at UBC to develop innovative solutions and contribute to the broader engineering community. I believe UBC's MEng program offers the right balance of depth and flexibility to help me grow in these areas. I'm particularly interested in coursework such as CPEN 511 (Computer Systems), EECE 550 (Power Electronics), ELEC 400M (Machine Learning), EECE 597 (Directed Research), and relevant lab-based electives that will allow me to apply theory directly to real world practice.

My decision to return to school is also influenced by the mentorship I have received throughout my life. My grandfather, my childhood neighbour, and my high school electronics teacher all instilled the value of curiosity and higher education. Their influence not only helped shape my technical path but also imparted the importance of giving back. I hope to continue that legacy by eventually mentoring younger students and engineers through work, volunteering, and academic outreach. I view the MEng not just as an opportunity to refine my technical skills, but as a platform to elevate my impact as an engineer, leader, and mentor.

In summary, I am applying to UBC's MEng in Electrical and Computer Engineering to strengthen my skills in embedded systems and power electronics, advance my career in technical design, and contribute to the engineering community through mentorship and service. With my background, curiosity, and clear goals, I am confident I will thrive and contribute to UBC's dynamic and hands-on learning environment.

Sincerely,

Navraj Singh Kambo